SCSI 1013: Discrete Structure

CHAPTER 6

FINITE AUTOMATA

2018/2019 – SEM. 1

Finite State Machines (FSM)

- Automata with input as well as output.
- Every state has an input and corresponding to the input the state also has an output.
- These types of automata are commonly called **finite state machines**.

Finite State Machines (FSM)

- A finite state machine is a sextuple,
  $M = \{ S, I, O, q_0, f_s, f_o \}$
  where,
  S is a finite nonempty set of states
  I is the input alphabet
  O is the output alphabet
  q_0 is the initial state
  f_s is the state transition function
  f_0 is the output function.

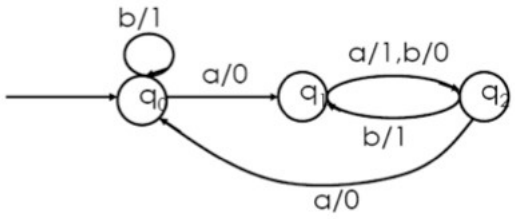
Example 1

- Let $M = \{ S, I, O, q_0, f_s, f_o \}$ be the FSM
- where,
  $S = \{q_0, q_1, q_2\}$,
  $I = \{a, b\}$,
  $O = \{0, 1\}$,
  q_0 = initial state,
- The f_s and f_0 are =>

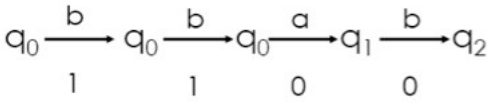
	f_s a	f_s b	f_o a	f_o b
q_0	q_1	q_0	0	1
q_1	q_2	q_2	1	0
q_2	q_0	q_1	0	1

Example 1 (cont.)

- The transition diagram:

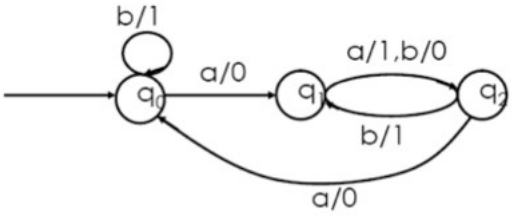


Input string: bbab

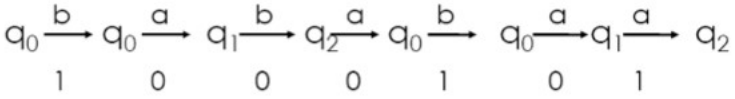


Output string: 1100 Output: 0

Example 1 (cont.)



Input string: bababaa



Output string: 1000101 Output: 1

Example 2

- Let M= { S, I, O, q_0, f_s, f_o} be the FSM

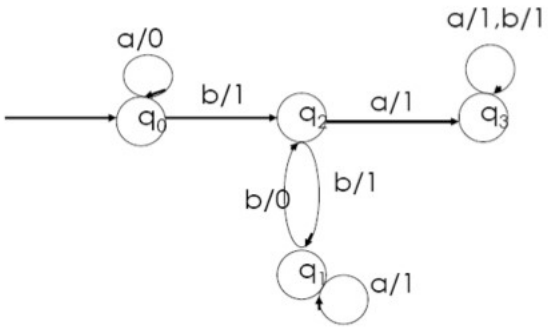
- where,
  S ={q_0,q_1,q_2 ,q_3},
  I ={a,b},
  O ={0,1},
  q_0 = initial state,

- $f_s$and f_0

	f_s a	b	f_o a	b
q_0	q_0	q_2	0	1
q_1	q_1	q_2	1	0
q_2	q_3	q_1	1	1
q_3	q_3	q_3	1	1

Example 2 (cont.)

- Draw the transition diagram of M.

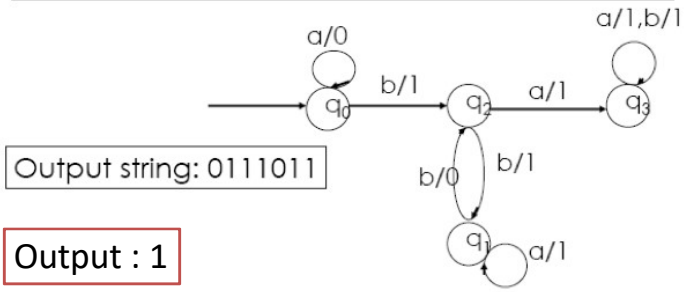


Example 2 (cont.)

- What is the output string if the input string is *abbabab*?

$q_0 \xrightarrow{a} q_0 \xrightarrow{b} q_2 \xrightarrow{b} q_1 \xrightarrow{a} q_1 \xrightarrow{b} q_2 \xrightarrow{a} q_3 \xrightarrow{b} q_3$

0 1 1 1 0 1 1



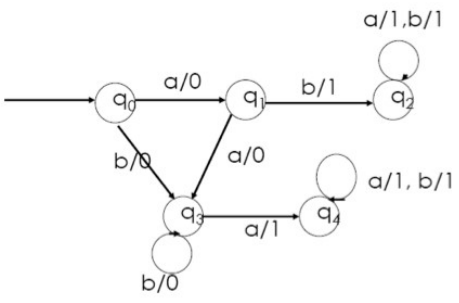
Output string: 0111011

Output : 1

Example 3

- Let M be a FSM.
- Let x be a nonempty string in M.
- We say that x is accepted by M if and only if the output of x is 1.

State transition diagram



Example 3 (cont.)

Based on the given information, answer the following:

- Write the transition table of M.
- What is the output string if the input string is *aaabbbb*?
- What is the output if the input string is *bbbaaaa*?
- Is the string *aaa* accepted by M?
- Which of the strings
  ba, aabbba, bbbb, aaabbbb
  are accepted by M?

Example 3 - Solution

- The transition table of M.

	f_s a	b	f_o a	b
q_0	q_1	q_3	0	0
q_1	q_3	q_2	0	1
q_2	q_2	q_2	1	1
q_3	q_4	q_3	1	0
q_4	q_4	q_4	1	1

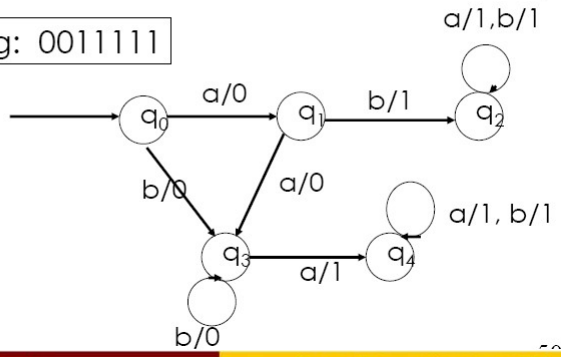
Solution (cont.)

- What is the output string if the input string is **aaabbbb**?

$q_0 \xrightarrow{a} q_1 \xrightarrow{a} q_3 \xrightarrow{a} q_4 \xrightarrow{b} q_4 \xrightarrow{b} q_4 \xrightarrow{b} q_4 \xrightarrow{b} q_4$

0 0 1 1 1 1 1

Output string: 0011111



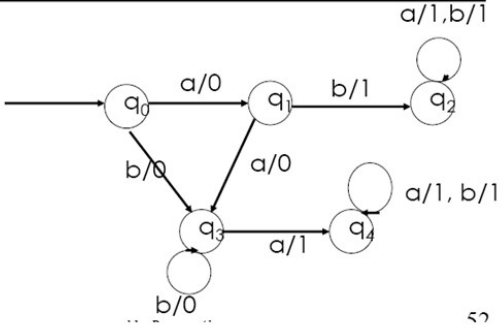
Solution (cont.)

- What is the output string if the input string is **bbbaaaa**?

$q_0 \xrightarrow{b} q_3 \xrightarrow{b} q_3 \xrightarrow{b} q_3 \xrightarrow{a} q_4 \xrightarrow{a} q_4 \xrightarrow{a} q_4 \xrightarrow{a} q_4$

0 0 0 1 1 1 1

Output: 1



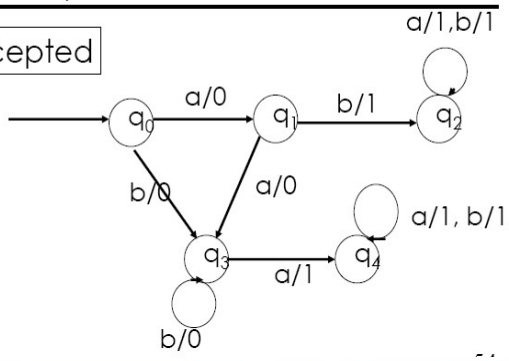
Solution (cont.)

- Is the string **aaa** accepted by M?

$q_0 \xrightarrow{a} q_1 \xrightarrow{a} q_3 \xrightarrow{a} q_4$

0 0 1

Output: 1,accepted



Solution (cont.)

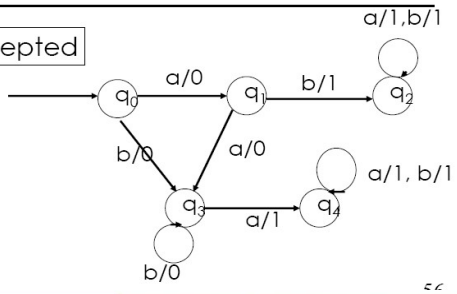
- Which of the strings **ba, aabbba, bbbb, aaabbbb** are accepted by M?

ba

$q_0 \xrightarrow{b} q_3 \xrightarrow{a} q_4$

0 1

Output: 1,accepted



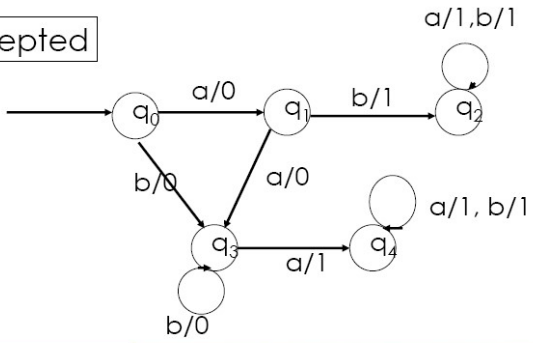
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aabbba

$q_0 \xrightarrow{a} q_1 \xrightarrow{a} q_3 \xrightarrow{b} q_3 \xrightarrow{b} q_3 \xrightarrow{b} q_3 \xrightarrow{a} q_4$

0 0 0 0 0 1

Output: 1,accepted



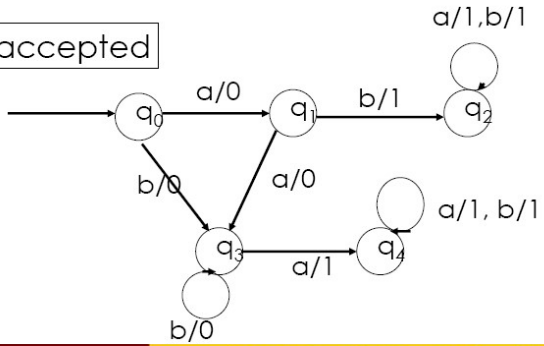
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bbbb

$q_0 \xrightarrow{b} q_3 \xrightarrow{b} q_3 \xrightarrow{b} q_3 \xrightarrow{b} q_3$

0 0 0 0

Output: 0,not accepted

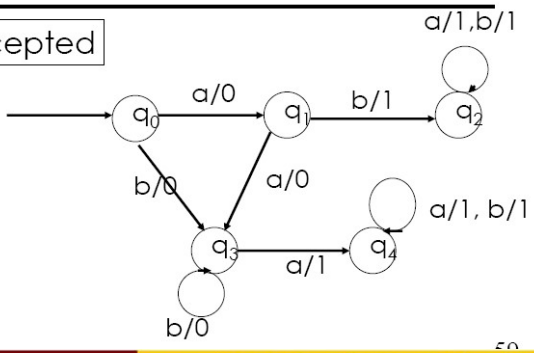


aaabbbb

$q_0 \xrightarrow{a} q_1 \xrightarrow{a} q_3 \xrightarrow{a} q_4 \xrightarrow{b} q_4 \xrightarrow{b} q_4 \xrightarrow{b} q_4 \xrightarrow{b} q_4$

0 0 1 1 1 1 1

Output: 1,accepted



Example 4

- Consider a vending machine that sells candy and the cost of a candy is 50 cents.
- The machine accepts any sequence of 10-, 20-, or 50 cent coins.
- After inserting at least 50 cents, the customer can press the button to release the candy.

- If the customer inputs more than 50 cents, the machine does not return the change.
- After selling the candy, the machine returns to initial state.
- Construct a finite state machine that models this vending machine.

Example 4 - Solution

1. List the states, S:

States,

q_0,	initial state (0)
q_1,	10 cents
q_2,	20 cents
q_3,	30 cents
q_4,	40 cents
q_5,	≥ 50 cents

2. List the elements of M:

$S = \{q_0, q_1, q_2, q_3, q_4, q_5\}$,

$I = \{10, 20, 50, B\}$,

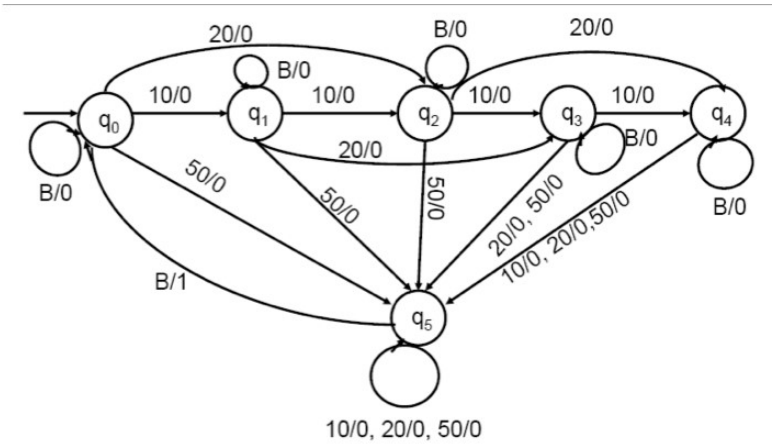
$O = \{0, 1\}$,

q_0 = initial state,

3. List the f_s and f_0:

	f_s				f_o			
	10	20	50	B	10	20	50	B
q_0	q_1	q_2	q_5	q_0	0	0	0	0
q_1	q_2	q_3	q_5	q_1	0	0	0	0
q_2	q_3	q_4	q_5	q_2	0	0	0	0
q_3	q_4	q_5	q_5	q_3	0	0	0	0
q_4	q_5	q_5	q_5	q_4	0	0	0	0
q_5	q_5	q_5	q_5	q_0	0	0	0	1

4. Draw the transition diagram of M:



Exercise #1

Let M= { S, I, O, q_0, f_s, f_o} be a FSM where,

S ={q_0,q_1,q_2},
I ={a,b},
O ={0,1},
q_0 = initial state,

f_s and f_0

	f_s a	f_s b	f_o a	f_o b
q_0	q_2	q_1	1	1
q_1	q_2	q_2	0	0
q_2	q_1	q_2	1	1

- Draw the transition diagram of M.
- What is the output string if the input string is *aabbb*?
- What is the output string if the input string is *ababab*?
- What is the output if the input string is *abbbaba*?
- What is the output if the input string is *bbbababa*?

Exercise #2

Let $M = (\{q_0, q_1, q_2, q_3, q_4, q_5\}, \{a, b, c\}, q_0, f_s, \{q_1, q_3, q_5\})$ be the Deterministic Finite Automaton (DFA) with state transition function, f_s defined as follows:

$f(q_0, a) = q_1$	$f(q_0, b) = q_0$	$f(q_0, c) = q_0$
$f(q_1, a) = q_1$	$f(q_1, b) = q_2$	$f(q_1, c) = q_1$
$f(q_2, a) = q_2$	$f(q_2, b) = q_3$	$f(q_2, c) = q_4$
$f(q_3, a) = q_3$	$f(q_3, b) = q_3$	$f(q_3, c) = q_3$
$f(q_4, a) = q_4$	$f(q_4, b) = q_5$	$f(q_4, c) = q_4$
$f(q_5, a) = q_5$	$f(q_5, b) = q_5$	$f(q_5, c) = q_5$

(a) Draw the transition table for the above machine.

(b) Determine the final state for the input string *abcc*.

(c) Is the input string *abcb* accepted by the DFA?

Exercise #3

Let $A = (S, I, O, Z, f, g)$ be a finite state machine (FSM) defined by the transition table shown in Table 1.

Table 1: Transition table of FSM *A*

Input / State	f			g		
	a	b	c	a	b	c
X	Z	X	Y	1	0	1
Y	X	X	Z	0	1	0
Z	Y	X	Z	1	0	1

(a) Draw the transition diagram of the finite state machine A.

(b) Find the output string for the input string *babccaab*.

(c) Find the output generated from the input string *cabcccba*.

(d) Determine whether the input string *abcbcbcabcc* is accepted.

Exercise #4

A sliding barrier turnstile (shown in Figure 1), used to control access to subways, is an automated gate at waist height with a barrier across the entryway. Initially the barrier is locked, barring the entry, preventing passengers from passing through. Depositing a token in a slot on the turnstile unlocks the barrier, allowing a single customer to push through. After the customer passes through, the barrier is locked again until another coin is inserted.



Figure 1

Considered as a state machine, the turnstile has two states: ***Locked*** and ***Unlocked***. There are two inputs that affect its state: putting a token in the slot (***token***) and retract the barrier (***retract***). In the locked state, retracting the arm has no effect; no matter how many times the input ***retract*** is given, it stays in the locked state. Putting a ***token*** as an input, shift the state from ***Locked*** to ***Unlocked***. In the unlocked state, putting additional tokens does not change the state. However, a customer passing through the retracted barrier, giving a ***retract*** input, shifts the state back to ***Locked***.

Given
- State
  - L: Locked
  - U: Unlocked
- Input
  - T: Token
  - R: Retract
- Output
  - 0: Nothing happened
  - 1: Retract the barrier so passenger can pass through
  - 2: Lock the barrier when passenger has passed through

(a) Complete the transition table below.

Table 1

State	Input, f_s		Output, f_o	
	T	R	T	R
L				
U				

(b) Draw the transition diagram for the turnstile system described above.